Introduction to Probability and Applied Statistics

Week 10: Linear Regression III

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Roadmap

Where we are

Omitted variable bias

Three routes to a zero conditional mean

Dummy variables

Interactions

Polynomials

Adjusted R^2

Assumption 4

The model:

$$Y = \beta_0 + \beta_1 X + \varepsilon.$$

ASSUMPTION 4: ZERO CONDITIONAL MEAN

$$\mathbb{E}[\varepsilon \mid X = x] = 0 \quad \text{for every } x.$$

- At every value of X, the average of ε is 0.
- If Assumptions 1–4 hold, OLS is unbiased: $\mathbb{E}[\hat{\beta}_1] = \beta_1$.

The country regression

We regressed under-5 mortality on government effectiveness, 46 sub-Saharan African countries:

$$\widehat{\texttt{u5mort}} = 40.9 - 31.6\,\texttt{goveff}.$$

- `u5mort`: deaths before age 5 per 1,000 live births.
- `goveff`: the World Bank's government effectiveness index.
- On average, a country with a one-unit higher `goveff` has 31.6 fewer under-5 deaths per 1,000 live births.

If Assumptions 1–4 hold, $\hat{\beta}_1$ is an unbiased estimate of β_1. Is Assumption 4 plausible for these countries?

Go BLUE or go home?

Some argue that the assumptions for OLS to be BLUE are **too restrictive**. For example:

- The target is the **best linear predictor** (BLP) of Y given X: the line that minimizes $\mathbb{E}[(Y - b_0 - b_1 X)^2]$. OLS is consistent for the BLP coefficients under iid sampling (plus variation in X). Linearity, zero conditional mean and homoskedasticity are not needed. (Aronow & Miller 2019)
- The population regression is the best linear approximation to the CEF. The approximation result holds even when the CEF is not linear. The classical assumptions are not needed for this justification. (Angrist & Pischke 2009)
- When the linear model is misspecified, which is the usual case, OLS estimates a well-defined approximation. Heteroskedasticity-robust standard errors remain valid. (Buja et al. 2019)

Go BLUE or go home? (continued)

1. **Descriptive target (this class).** OLS consistently estimates the BLP. We need iid sampling and variation in X.
   - A4 is not needed, though A4 is useful for thinking about "what's missing" from our regression.
   - A5 is not needed as long as we estimate heteroskedasticity-robust standard errors (later in the course).
2. **Causal target (causal inference).** When is a BLP slope equal to a causal effect? When Y follows a linear causal model, $Y = \beta_0 + \beta_1 X + \varepsilon$, and A4 holds for that model's ε. Research design is how we make A4 plausible.

tl;dr: the classical assumptions describe the ideal case, not the conditions for regression to be useful.

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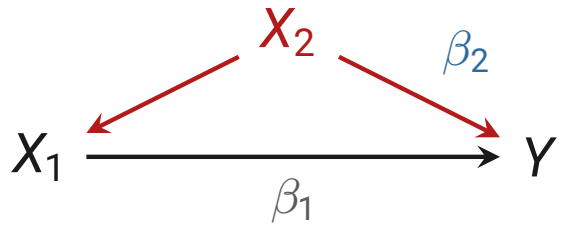
Polynomials

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An assumption we cannot check from the data

- OLS is unbiased if Assumptions 1–4 hold.
- We can check A3 (variation in X / no perfect collinearity) in the sample. A1 (linearity) concerns the form of the model and A2 (random sampling) how we drew the sample.
- A4 is a claim about ε, and we never observe ε.
- The most common failure of A4: ε contains a variable that affects Y and is correlated with X.

Suppose X_2 is a second variable that affects Y and is correlated with X_1. What does $\hat{\beta}_1$ estimate?

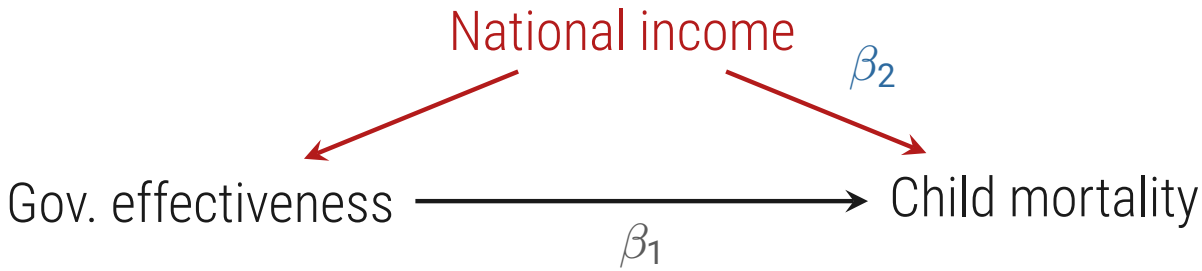


On average, $\hat{\beta}_1$ (the slope we estimate without X_2 in our regression) equals β_1 plus a second term.

An omitted variable in the country data

- We regress under-5 mortality on government effectiveness in 46 sub-Saharan African countries. $\hat{\beta}_1 < 0$: countries with more effective governments have lower child mortality.
- Richer countries have more effective governments. Richer countries also have lower child mortality.
- National income is not in the regression, so national income is part of ε.

Does A4 hold if we don't account for national income? Is the average of ε the same in countries with effective and ineffective governments?



Why not include X_2?

We often cannot add X_2 to the regression:

- We are reading someone else's analysis.
- X_2 is not in our data.
- We do not know which variables to add.

We may not be able to remove the bias. We can often work out its sign.

The long and the short regression

Suppose the **long regression** is the causal model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X_1, X_2] = 0.$$

β_1 is the change in average Y when X_1 rises by one unit and X_2 stays the same ("controlling for X_2").

$\mathbb{E}[\varepsilon \mid X_1, X_2] = 0$ means A4 holds for the long regression: for every combination of X_1 and X_2, the average of ε is 0.

But, suppose we do not observe X_2, so we fit the **short regression**, Y on X_1 alone:

$$Y = \beta_0^* + \beta_1^* X_1 + v.$$

Call its estimated slope $\hat{\beta}_1^*$.

We want β_1. The short regression gives us $\hat{\beta}_1^*$.

$$\text{Bias} = \mathbb{E}[\hat{\beta}_1^*] - \beta_1.$$

While we cannot eliminate this bias, we can do a bit of math to understand how it affects our estimates.

Step 1: regress the omitted variable on the included one

We start by estimating the relationship between X_1 (the variable we do observe) and X_2 (the omitted variable):

The **auxiliary regression**:

$$X_2 = \delta_0 + \delta_1 X_1 + w, \qquad \mathbb{E}[w \mid X_1] = 0.$$

- δ_1 is the population slope from regressing X_2 on X_1: the change in the average of X_2 when X_1 rises by one unit. If we observed X_2, we could estimate δ_1.
- $\delta_1 = \mathrm{Cov}(X_1, X_2)/\mathbb{V}[X_1]$, so δ_1 has the sign of the correlation between X_1 and X_2.

Steps 2–4: substitute and group

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

The long regression.

$$= \beta_0 + \beta_1 X_1 + \beta_2 (\delta_0 + \delta_1 X_1 + w) + \varepsilon$$

Step 2. Replace X_2 with the auxiliary regression.

With X_2 written in terms of X_1, the equation contains Y, X_1 and the error terms.

$$= \beta_0 + \beta_1 X_1 + \beta_2 \delta_0 + \beta_2 \delta_1 X_1 + \beta_2 w + \varepsilon$$

Step 3. Multiply β_2 through the parentheses.

$$= \underbrace{(\beta_0 + \beta_2 \delta_0)}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2 \delta_1)}_{\beta_1^*} X_1 + \underbrace{(\beta_2 w + \varepsilon)}_{v}$$

Step 4. Group the constants, the terms in X_1 and the error terms.

The last line has the form of the short regression: a constant, a slope times X_1, and an error.

Steps 5–6: the bias

Step 5. The error $\beta_2 w + \varepsilon$ has mean 0 at every value of X_1, because w and ε both do. So the last line (Step 4) *is* the short regression, $Y = \beta_0^* + \beta_1^* X_1 + v$. Its slope is

$$\beta_1^* = \beta_1 + \beta_2 \, \delta_1.$$

- β_2: the slope on X_2 in the long regression (omitted variable $\rightarrow$ outcome).
- δ_1: the slope from regressing X_2 on X_1 (included variable $\rightarrow$ omitted variable).

STEP 6. OMITTED VARIABLE BIAS

$$\text{Bias} = \mathbb{E}[\hat{\beta}_1^*] - \beta_1 = \beta_1^* - \beta_1 = \beta_2 \, \delta_1.$$

Step 5 shows that v has mean 0 at every value of X_1. So $\mathbb{E}[\hat{\beta}_1^*] = \beta_1^*$.

When does omitting X_2 matter?

$$\mathbb{E}[\hat{\beta}_1^*] - \beta_1 = \beta_2 \, \delta_1$$

Omitting X_2 makes β_1^* differ from β_1 only if both hold:

1. $\beta_2 \neq 0$: X_2 is related to Y, holding X_1 fixed;
2. $\delta_1 \neq 0$: X_2 is correlated with X_1.

Both conditions hold for a **confounder**: a variable that affects both X_1 and Y. For a confounder, $\beta_2 \delta_1$ is bias.

A **mediator** also meets both conditions: a variable that X_1 affects and that affects Y. For a mediator, $\beta_2 \delta_1$ is the part of the effect of X_1 that runs through X_2.

The sign of the bias in $\hat{\beta}_1^*$

Bias = $\beta_2 \, \delta_1$. Since $\delta_1 = \text{Cov}(X_1, X_2)/\mathbb{V}[X_1]$ and $\mathbb{V}[X_1] > 0$, δ_1 has the same sign as $\text{Cov}(X_1, X_2)$. So the sign of the bias depends on the sign of β_2 and the sign of $\text{Cov}(X_1, X_2)$.

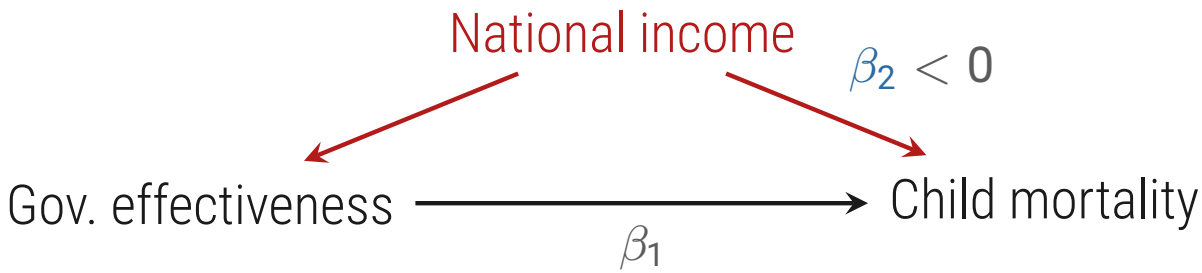
	$\text{Cov}(X_1, X_2) > 0$	$\text{Cov}(X_1, X_2) < 0$
$\beta_2 > 0$	positive bias	negative bias
$\beta_2 < 0$	negative bias	positive bias

Positive bias: $\mathbb{E}[\hat{\beta}_1^*] > \beta_1$. Negative bias: $\mathbb{E}[\hat{\beta}_1^*] < \beta_1$.

Back to the country data

Omitted variable: national income.

- $\beta_2 < 0$: richer countries have lower child mortality, holding government effectiveness fixed.
- $\delta_1 > 0$: richer countries have more effective governments.

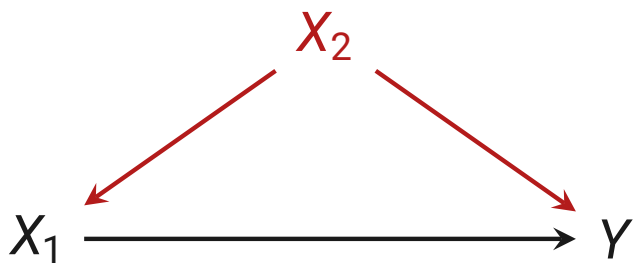


Bias = $\beta_2\,\delta_1 < 0$. So $\beta_1^* < \beta_1$: on average, $\hat{\beta}_1^*$ is less than β_1.

Careful! When we say that the true effect is greater than (i.e., *less negative* than) our estimate, this means that the true effect could be negative, zero or positive.

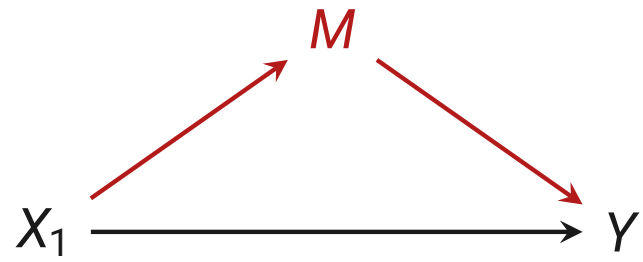
Confounders and mediators

Confounder



Failing to control for X_2 biases $\hat{\beta}_1$.

Mediator



Do not control for M.

Which variables do we control for?

X_2 is a ...	**confounder**	**mediator**
Arrows	$X_1 \leftarrow X_2 \rightarrow Y$	$X_1 \rightarrow X_2 \rightarrow Y$
Short slope β_1^*	the effect of X_1 mixed with the effect of X_2	the total effect of X_1
Long slope β_1	the effect of X_1	the direct effect of X_1 only
$\beta_2\delta_1$ is ...	bias	the part of the effect of X_1 that runs through X_2
Control for X_2?	Yes	No, if we want the total effect

A mediator in the country data: government effectiveness $\rightarrow$ health spending $\rightarrow$ child mortality. To estimate the total effect of government effectiveness, we leave health spending out of the regression.

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The problem

How can we make $\mathbb{E}[\varepsilon \mid X] = 0$ plausible?

1. **Controls.** Measure every common cause of X and Y and include each one in the regression.
2. **Research design.** Find cases where X varies for reasons unrelated to the other influences on Y.
3. **Random assignment.** Set X ourselves, at random.

The coefficient of interest and the controls

$$Y_i = \beta_0 + \beta_1 X_i + \gamma W_i + \varepsilon_i$$

- β_1: the coefficient we care about.
- W_i: one or more control variables. γ: their coefficients.

If $\mathbb{E}[\varepsilon \mid X, W] = 0$, OLS is unbiased for every coefficient.

Often we want an unbiased estimate of β_1 only. We include the controls so that $\hat{\beta}_1$ is unbiased. The coefficients on the controls need not be causal effects.

Route 1: controls

Does campaign contact raise turnout?

2024 American National Election Study: "Has anyone from one of the political parties called you up or come around and talked to you about the 2024 campaign . . . ?"

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{contacted}_i + \gamma W_i + \varepsilon_i$$

Campaigns contact people they expect to vote. Variables that might be related to turnout and correlated with contact:

- interest in politics; turnout in past elections
- age; education
- party registration

Problem: how do we know we have controlled for every confounder?

Route 2: research design

Does campaign contact raise turnout?

Enos and Fowler (2018): media markets that cross state lines, 2012 election.

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{battleground}_i + \varepsilon_i, \qquad \text{within one media market}$$

- Presidential campaigns concentrate on battleground states. Some media markets straddle the border between a battleground state and a non-battleground state.
- Voters on the two sides of the border watch the same local TV. Campaign contact differs because of the Electoral College.
- The authors check: the two sides had similar demographics and similar turnout in 2010, an election with no presidential campaign.

β_1 is the effect of living on the battleground side of the border. Both sides see the same TV ads. The battleground side gets more canvassing, phone calls and mail.

Problem: do the two sides of a state boundary differ only in campaign contact?

Route 3: random assignment

Does campaign contact raise turnout?

A field experiment in New Haven, 1998 (Gerber and Green 2000):

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{canvass}_i + \varepsilon_i$$

- Registered voters were assigned at random to a door-to-door canvassing visit or to no visit: every voter had the same probability of assignment.
- Then $\mathbb{E}[\varepsilon \mid \texttt{canvass}] = 0$: averaged over every possible random assignment, the two groups have the same average ε. Gerber and Green assigned voters at random only once. Like one random sample, one random assignment can produce groups that differ by chance on parts of ε. Such chance differences are sampling error, not bias. The SE measures their typical size.

β_1 is the effect of being assigned to a visit. (We call this an "intent to treat" effect because canvassers reached only some of the assigned voters.)

Problem: which interesting variables can we assign at random?

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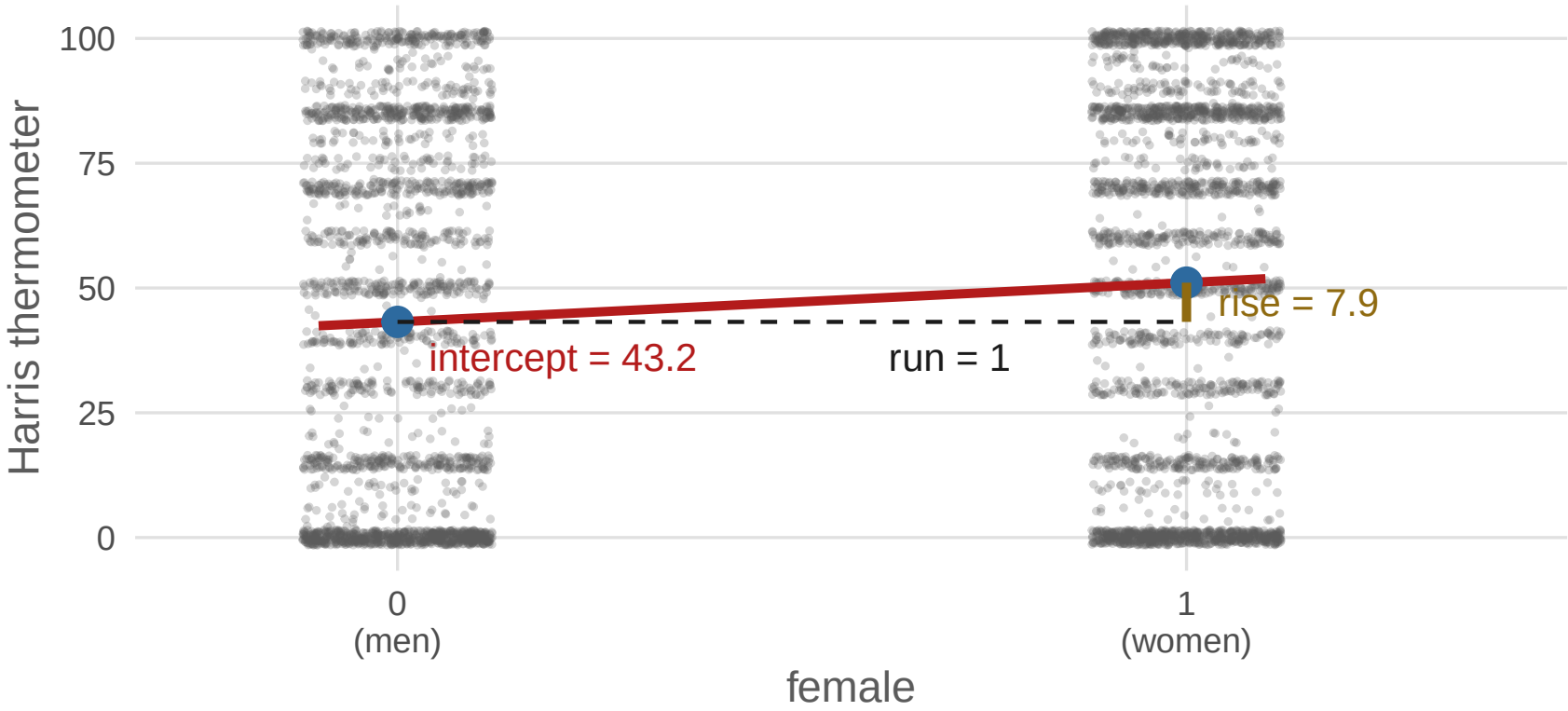
A variable coded 1 for one group and 0 for the other is a **binary**, **dummy** or **indicator** variable.

id	harris	sex	female	educ	college	race	white
140001	0	Male	0	Bachelor's	1	Hispanic	0
140002	50	Female	1	Some college	0	Asian/NHPI	0
140003	90	Female	1	Some college	0	White	1
140004	50	Male	0	Some college	0	Asian/NHPI	0
140005	5	Male	0	Graduate	1	White	1
140006	85	Female	1	Some college	0	White	1

Variable	Equals 1 if the respondent ...	Share with 1
`female`	is female	0.54
`college`	has a bachelor's degree or more	0.45
`white`	is white, non-Hispanic	0.73

A line through two group means (revisited)

Y is the Harris feeling thermometer (0–100). n = 5,362.



In the bivariate case, when X takes only the values 0 and 1, OLS gives

$$\text{intercept} = \bar{Y}_{X=0}, \qquad \text{slope} = \bar{Y}_{X=1} - \bar{Y}_{X=0}.$$

Bivariate regressions with one dummy variable

group means

Regression	$\bar{Y}_{X=0}$	$\bar{Y}_{X=1}$	intercept	slope
`lm(harris ~ female, data = d3)`	43.2	51.1	43.2	7.9
`lm(harris ~ college, data = d3)`	41.6	54.4	41.6	12.8
`lm(harris ~ white, data = d3)`	58.4	43.3	58.4	−15.2

All three regressions use the same 5,362 respondents. Each of them answered the Harris, sex, education and race items. The data frame `d3` holds these respondents.

In the last row, the slope does not equal $\bar{Y}_{X=1} - \bar{Y}_{X=0}$ exactly because of rounding.

Do college graduates rate Harris higher than non-graduates of the same sex and race? To answer, we put all three dummies in one regression.

Three dummies in one regression

```
fit3 <- lm(harris ~ female + college + white, data = d3)
coef(fit3)
```

```
(Intercept)      female     college       white
      48.69        7.91       14.43      -16.58
```

$$\widehat{\texttt{harris}} = 48.7 + 7.9\,\texttt{female} + 14.4\,\texttt{college} - 16.6\,\texttt{white}$$

- `college`: on average, controlling for sex and race, respondents with a degree rate Harris 14.4 points higher than respondents without one.
- `female`: on average, controlling for education and race, women rate Harris 7.9 points higher than men.
- `white`: on average, controlling for sex and education, white respondents rate Harris 16.6 points lower than other respondents.
- Intercept: the predicted rating when every dummy equals 0. Every dummy equals 0 for a man without a degree who is not white: 48.7.

Predicted values

$$\widehat{\texttt{harris}} = 48.7 + 7.9\,\texttt{female} + 14.4\,\texttt{college} - 16.6\,\texttt{white}$$

To predict, plug in 0 or 1 for each dummy.

A white woman with a degree:

$$48.7 + 7.9(1) + 14.4(1) - 16.6(1) = 54.4$$

A white man without a degree:

$$48.7 + 7.9(0) + 14.4(0) - 16.6(1) = 32.1$$

A non-white woman without a degree:

$$48.7 + 7.9(1) + 14.4(0) - 16.6(0) = 56.6$$

All eight groups

Race	Education	Sex	*n*	Predicted	Group mean
not white	no degree	male	404	48.7	51.6
not white	no degree	female	518	56.6	59.3
not white	degree	male	252	63.1	57.0
not white	degree	female	296	71.0	67.6
white	no degree	male	922	32.1	31.7
white	no degree	female	1,090	40.0	38.0
white	degree	male	904	46.5	47.3
white	degree	female	976	54.4	56.3

The predictions are close to the group means, but not equal to them.

In this regression, the predicted values for the two values of a dummy differ by that dummy's coefficient. The difference is the same in every group defined by the other two dummies. For example, degree and no degree differ by 14.4 points in all four sex-by-race groups.

Categorical variables

Race in five categories (`race5`):

Code	Race	*n*	Mean Harris rating
1	White	3,934	43.2
2	Black	506	73.5
3	Hispanic	578	52.2
4	Asian/NHPI	197	49.7
5	Other or multiple	220	46.7

`race5` runs from 1 to 5, but the codes are labels. The categories have no order. A slope for "one more unit of `race5`" has no meaning.

One dummy per category, minus one

We make a dummy for each category except one, the **reference category**:

$$\widehat{\texttt{harris}} = \hat{\beta}_0 + \hat{\beta}_1 D_{\text{Black}} + \hat{\beta}_2 D_{\text{Hispanic}} + \hat{\beta}_3 D_{\text{Asian/NHPI}} + \hat{\beta}_4 D_{\text{other}}$$

- Reference category: White. A white respondent has every $D = 0$.
- Why leave one out? Every respondent is in exactly one category. The five dummies therefore sum to 1 for every respondent.
- The intercept column also equals 1 for every respondent. With all five dummies, one column equals an exact linear function of the others. Assumption 3 then fails.

In R: `factor()`

```
lm(harris ~ factor(race5), data = anes)
```

```
(Intercept)        Black     Hispanic  Asian/NHPI        Other
      43.21        30.33         8.95         6.46         3.53
```

- Intercept: the mean for the reference category, White.
- Each slope: that category's mean minus the reference mean. For Black respondents:

$$\underbrace{73.54}_{\substack{\text{mean Harris rating} \\ \text{for Black respondents}}} - \underbrace{43.21}_{\substack{\text{intercept} \\ \text{(reference category)}}} = \underbrace{30.33}_{\substack{\text{slope on the} \\ \text{Black dummy variable}}}$$

Changing the reference category

```
lm(harris ~ relevel(factor(race5), ref = "2"), data = anes)
```

Race	Slope, reference = White	Slope, reference = Black
White	reference	−30.3
Black	30.3	reference
Hispanic	9.0	−21.4
Asian/NHPI	6.5	−23.9
Other or multiple	3.5	−26.8

Some slopes differ by 0.1 from the difference between the rounded means because of rounding.

- Each slope is a difference from the reference category. A new reference category gives new slopes.
- The fitted values do not change. Each respondent's prediction is the mean of their category in both regressions.

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Is the college difference the same for every group?

Mean Harris rating:

	No degree	Degree	Degree — no degree
Not white	55.9	62.7	6.8
White	35.1	52.0	16.9

In the data (not modeled), the college difference among white respondents is about 2.5 times the college difference among other respondents.

One difference for everyone

$$\widehat{\texttt{harris}} = 53.1 + 14.3\,\texttt{college} - 16.7\,\texttt{white}$$

Predicted	No degree	Degree	College difference
Not white	53.1	67.4	14.3
White	36.4	50.7	14.3

In a regression with `college` and `white` only, the predicted college difference is 14.3 for both groups. To let the gap differ, we add a new variable.

An interaction term

The **interaction** is the product of the two dummies:

$$\texttt{college} \times \texttt{white} = \begin{cases} 1 & \text{a white respondent with a degree} \\ 0 & \text{everyone else} \end{cases}$$

The model:

$$Y = \beta_0 + \beta_1 \texttt{college} + \beta_2 \texttt{white} + \beta_3 (\texttt{college} \times \texttt{white}) + \varepsilon$$

Interpreting interaction terms

$$Y = \beta_0 + \beta_1 \texttt{college} + \beta_2 \texttt{white} + \beta_3 (\texttt{college} \times \texttt{white}) + \varepsilon$$

	No degree	Degree
Not white	β_0	$\beta_0 + \beta_1$
White	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

- College difference, not white: $(\beta_0 + \beta_1) - \beta_0 = \beta_1$.
- College difference, white: $(\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) = \beta_1 + \beta_3$.
- β_3 is the difference between the two college differences.

Interaction terms in R

```
fit_int  <- lm(harris ~ college * white, data = d3)
fit_int2 <- lm(harris ~ college + white + college:white, data = d3)
rbind(coef(fit_int), coef(fit_int2))
```

```
     (Intercept) college white college:white
[1,]       55.91     6.8 -20.8         10.09
[2,]       55.91     6.8 -20.8         10.09
```

`college * white` is shorthand for `college + white + college:white`. The two regressions are the same.

Both regressions use the same 5,362 respondents. Each of them answered the Harris, sex, education and race items.

Always include `college` and `white` on their own. Leaving them out is the same as setting $\beta_1 = \beta_2 = 0$. Three of the four groups then have the same predicted value, β_0.

Four fitted values, four group means

$$\widehat{\texttt{harris}} = 55.9 + 6.8\,\texttt{college} - 20.8\,\texttt{white} + 10.1\,(\texttt{college} \times \texttt{white})$$

Fitted	No degree	Degree
Not white	55.9	62.7
White	35.1	52.0

With four groups and four coefficients, the fitted values equal the four group means.

Reading the coefficients

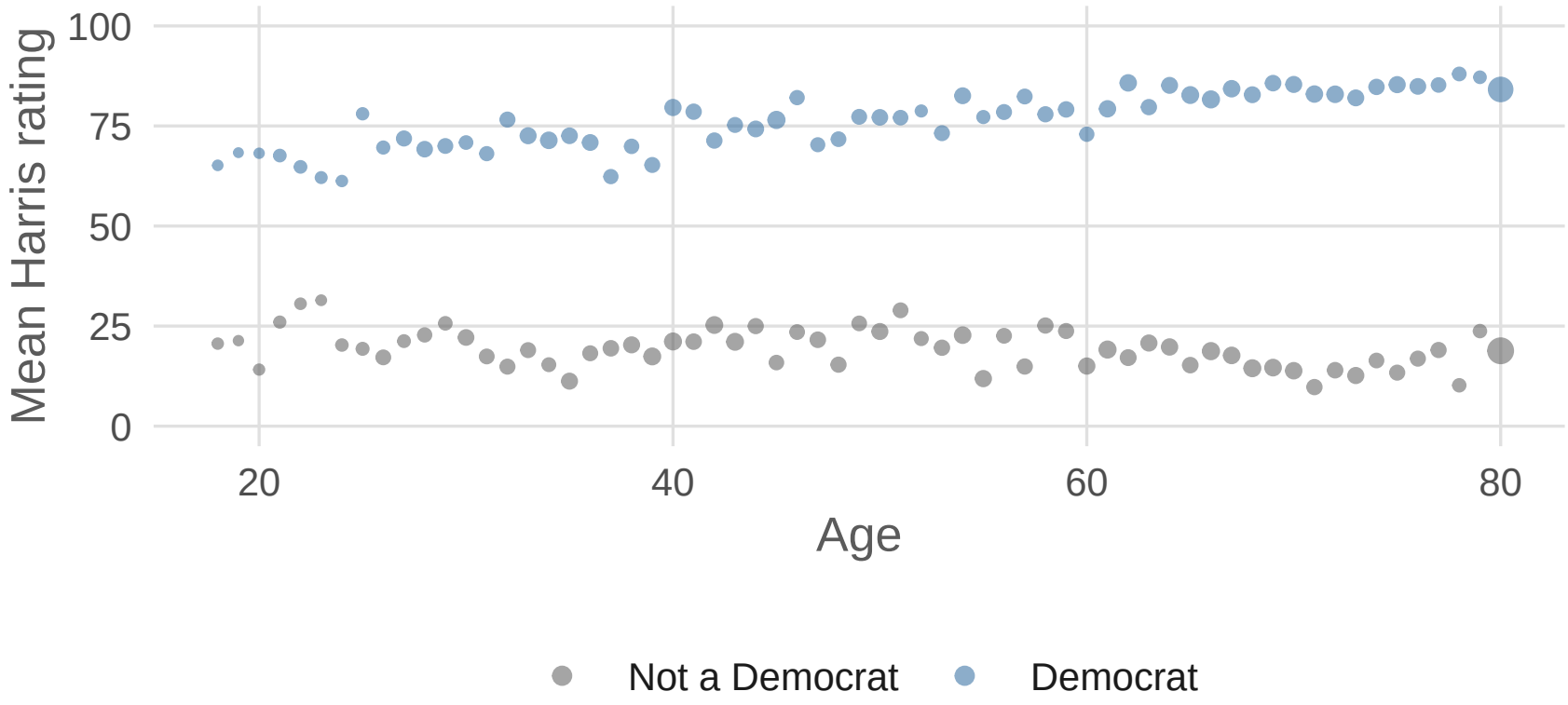
$$\widehat{\texttt{harris}} = 55.9 + 6.8\,\texttt{college} - 20.8\,\texttt{white} + 10.1\,(\texttt{college} \times \texttt{white})$$

- $\hat{\beta}_1 = 6.8$: among respondents who are not white, those with a degree rate Harris 6.8 points higher on average than those without one.
- $\hat{\beta}_1 + \hat{\beta}_3 = 6.8 + 10.1 = 16.9$: among white respondents, those with a degree rate Harris 16.9 points higher on average than those without one.
- $\hat{\beta}_3 = 10.1$: the difference between the two college differences.
- $\hat{\beta}_2 = -20.8$: among respondents **without a degree**, white respondents rate Harris 20.8 points lower on average than other respondents. Among respondents with a degree, the difference is $\hat{\beta}_2 + \hat{\beta}_3 = -10.7$.

In a model with an interaction, $\hat{\beta}_1$ is not "the" college difference. The college difference depends on race.

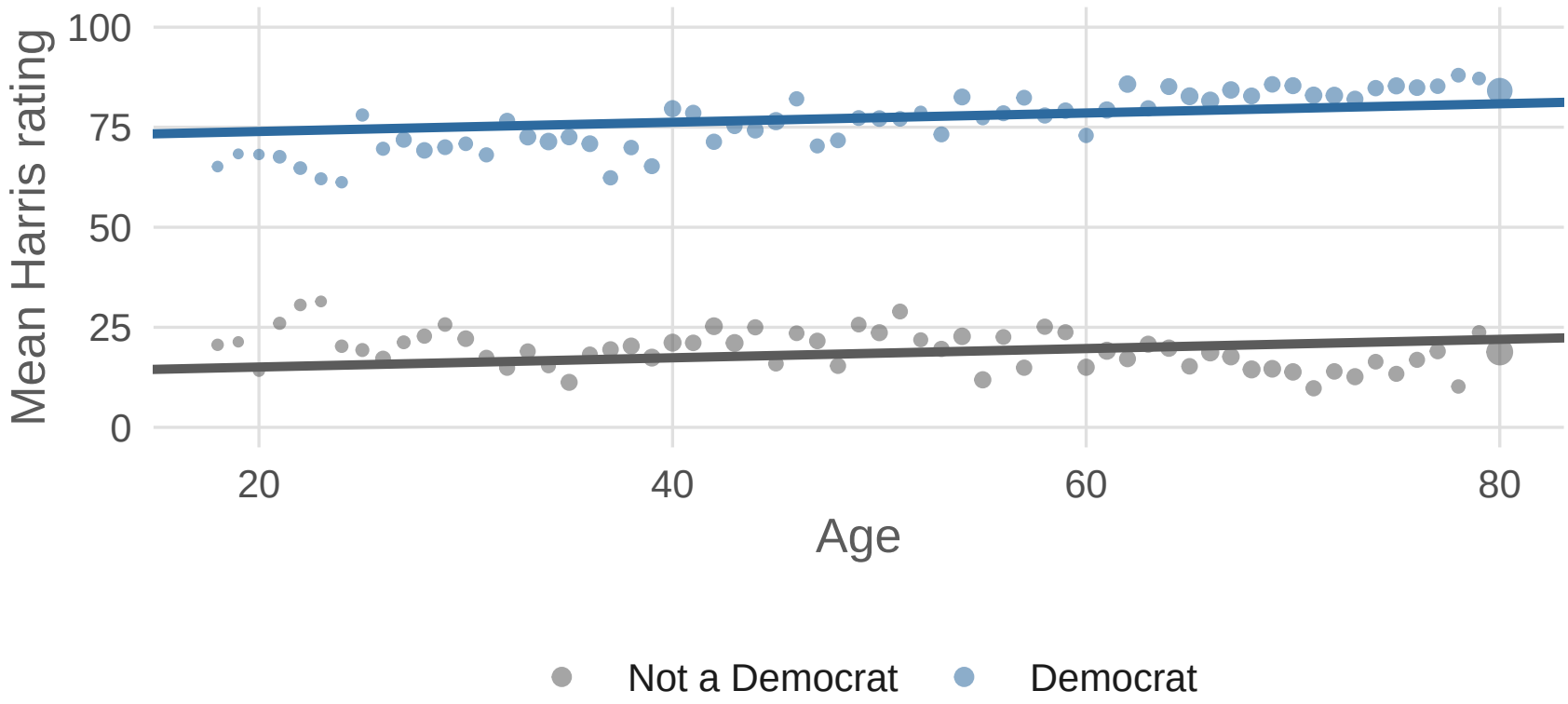
A dummy and a continuous variable

Mean Harris rating at each age, Democrats (leaners included) and everyone else:



Without an interaction: parallel lines

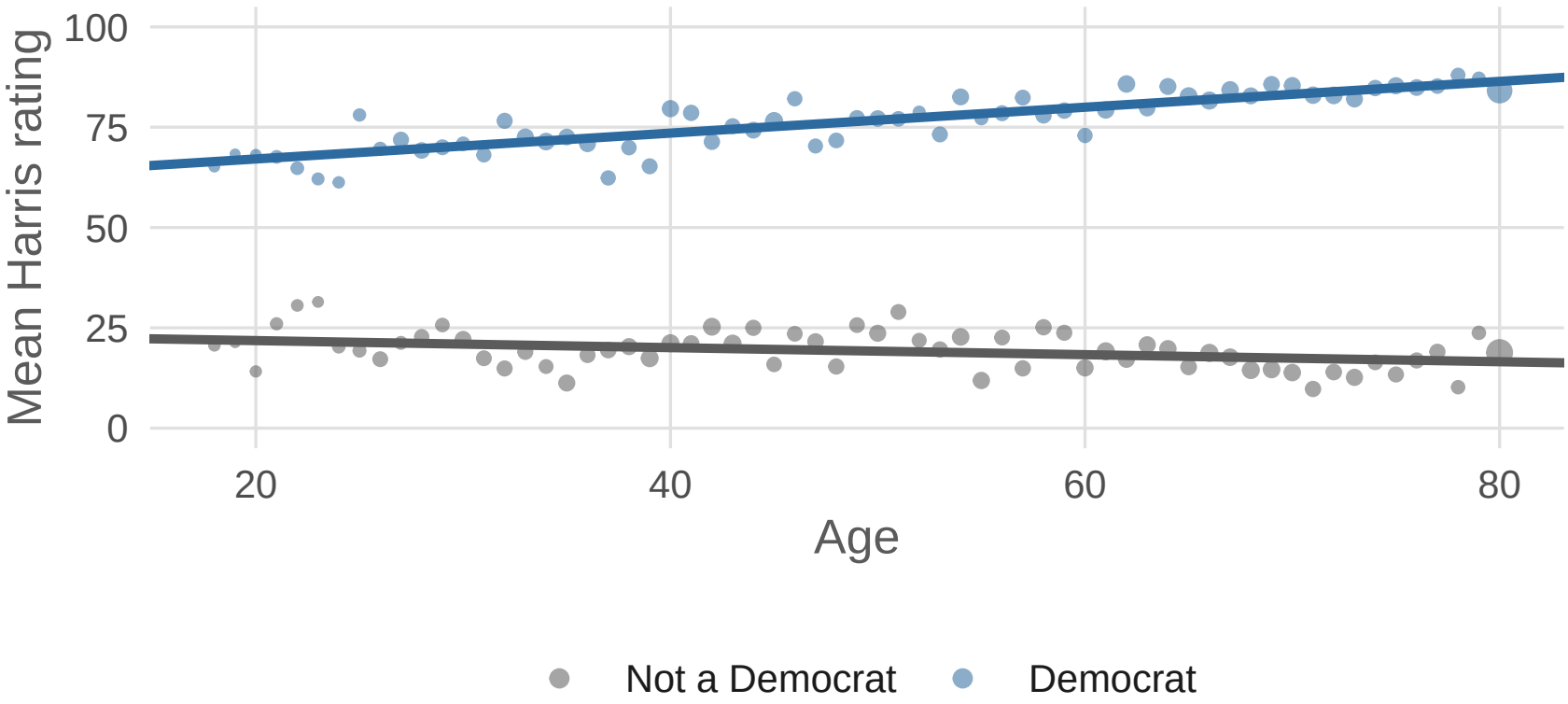
$$\widehat{\texttt{harris}} = 12.8 + 0.12\,\texttt{age} + 58.9\,\texttt{dem}$$



One slope on age for both groups: 0.12 points per year.

With an interaction: two (different) slopes

$$\widehat{\texttt{harris}} = 23.6 - 0.09\,\texttt{age} + 37.1\,\texttt{dem} + 0.41\,(\texttt{age} \times \texttt{dem})$$



Interpreting regressions with interaction terms

Set `dem` = 0:

$$\widehat{\texttt{harris}} = 23.6 - 0.09\,\texttt{age}$$

Set `dem` = 1:

$$\widehat{\texttt{harris}} = (23.6 + 37.1) + (-0.09 + 0.41)\,\texttt{age} = 60.7 + 0.32\,\texttt{age}$$

- Among non-Democrats, each year of age is associated with a rating 0.09 points lower, on average.
- Among Democrats, each year of age is associated with a rating 0.32 points higher, on average.
- The coefficient on `age:dem`, 0.41, is the difference between the two slopes.

Interpreting regressions with interaction terms (continued)

How to interpret the coefficient on `dem`?

$$\widehat{\texttt{harris}} = 23.6 - 0.09\,\texttt{age} + 37.1\,\texttt{dem} + 0.41\,(\texttt{age} \times \texttt{dem})$$

$\hat{\beta}_{\texttt{dem}} = 37.1$ is the difference between the two lines where `age` = 0.

- No respondent is 0 years old. The youngest respondents are 18.
- The difference between the lines changes with age:

$$\text{difference at age } a = 37.1 + 0.41\,a.$$

Age	25	50	75
Democrat minus non-Democrat	47.3	57.6	67.8

Marginal effects

The **marginal effect** of a variable is the change in $\hat{Y}$ when that variable rises by one unit, holding the other variables fixed.

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \texttt{age} + \hat{\beta}_2 \texttt{dem} + \hat{\beta}_3 (\texttt{age} \times \texttt{dem})$$

- Marginal effect of `age`: $\hat{\beta}_1 + \hat{\beta}_3 \texttt{dem}$. The marginal effect of `age` depends on `dem`.
- Marginal effect of `dem` (0 to 1): $\hat{\beta}_2 + \hat{\beta}_3 \texttt{age}$. The marginal effect of `dem` depends on `age`.

Note: the term "marginal effects" comes from economics and should not be confused with "causal effects". A marginal effect is a **partial derivative of the fitted regression**: the change in $\hat{Y}$ when one variable changes, holding the other variables in the model constant. When the model matches the CEF, marginal effects equal the partial derivatives of the CEF.

Re-centering in R

```
fit <- lm(harris ~ age * dem, data = anes)
round(coef(fit), 2)
```

```
(Intercept)         age         dem     age:dem
      23.57       -0.09       37.06        0.41
```

```
mean_age <- mean(anes$age, na.rm = TRUE)
mean_age
```

```
[1] 53.15052
```

```
anes$age_c <- anes$age - mean_age
fit_c <- lm(harris ~ age_c * dem, data = anes)
round(coef(fit_c), 2)
```

```
(Intercept)       age_c         dem   age_c:dem
      18.91       -0.09       58.86        0.41
```

58.86 = 37.06 (coefficient on **dem**) + 0.41 (coefficient on **age:dem**) $\times$ 53.15 (mean age)

Working with interaction terms

1. Include the **lower-order terms**. In a model with `x` $\times$ `z`, the lower-order terms are `x` and `z` on their own. In R, `x * z` adds `x`, `z` and `x:z`.
2. The coefficient on `x` is the marginal effect of `x` when `z` = 0. If we "re-center", i.e., subtract the mean of `z` from `z` before fitting, the coefficient on `x` equals the marginal effect at the mean of `z`.
3. Report marginal effects at values of `z` inside the range of the data. Good choices are the mean of `z` or a few typical values.

Marginal effects in R: `slopes()`

The marginal effect of `age`, for each value of `dem`:

```
library(marginaleffects)
slopes(fit, variables = "age", by = "dem")
```

```
 dem Estimate Std. Error     z Pr(>|z|)     S  2.5 %  97.5 %
   0  -0.0877     0.0257 -3.42   <0.001  10.6 -0.138 -0.0374
   1   0.3223     0.0259 12.43   <0.001 115.5  0.272  0.3731
```

- Row `dem` = 0: among non-Democrats, each additional year of age is associated with a predicted Harris rating 0.088 points lower, on average. This number equals the coefficient on `age` in `lm()`.
- Row `dem` = 1: among Democrats, each additional year of age is associated with a predicted rating 0.322 points higher, on average. This number equals the coefficient on `age` plus the coefficient on `age:dem`: $-0.088 + 0.410 = 0.322$.

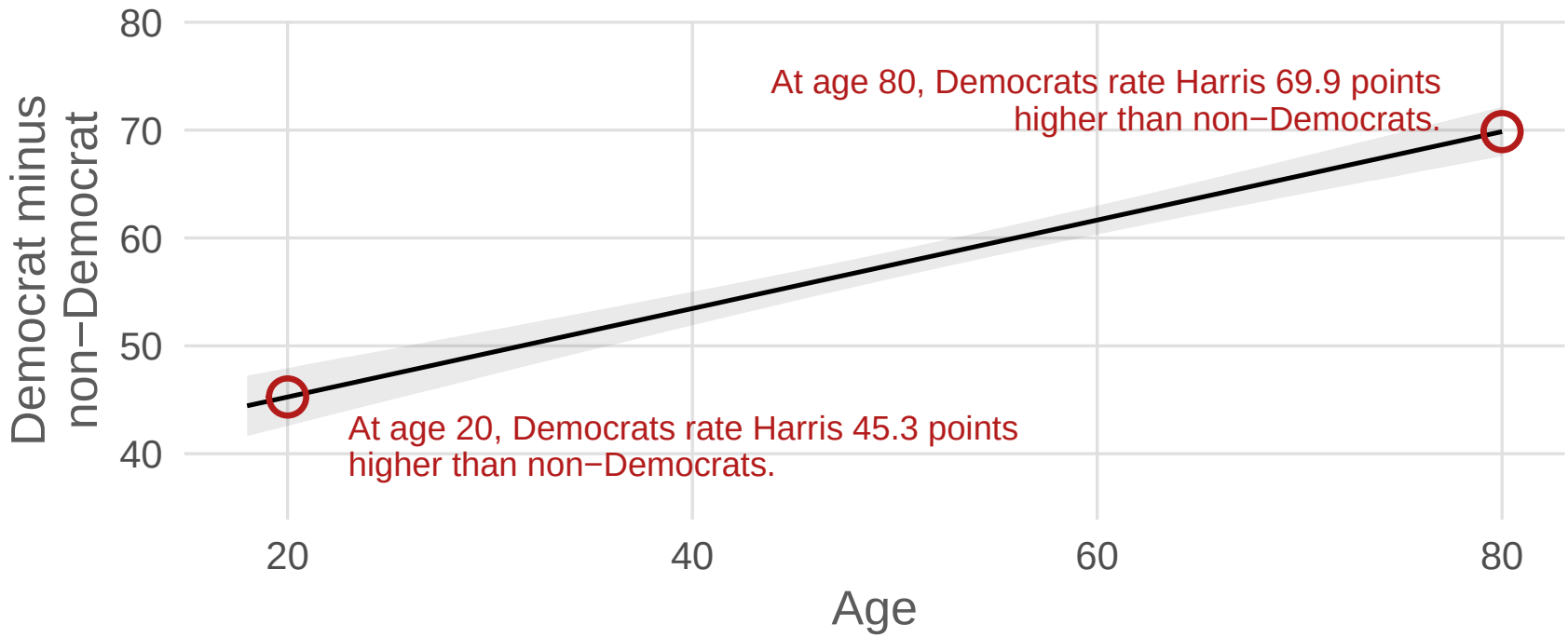
We derived these by hand a few slides back.

The function also returns a standard error for the Democrats' slope, which requires the covariance of the two coefficients.

Marginal effects in R: `plot_comparisons()`

The marginal effect of **`dem`** (0 to 1), at each age:

```
plot_comparisons(fit, variables = "dem", condition = "age")
```



Note: to check whether the gap changes with age along a straight line, use the binning estimator in the **`interflex`** package (Hainmueller, Mummolo and Xu 2019).

Roadmap

Where we are

Omitted variable bias

Three routes to a zero conditional mean

Dummy variables

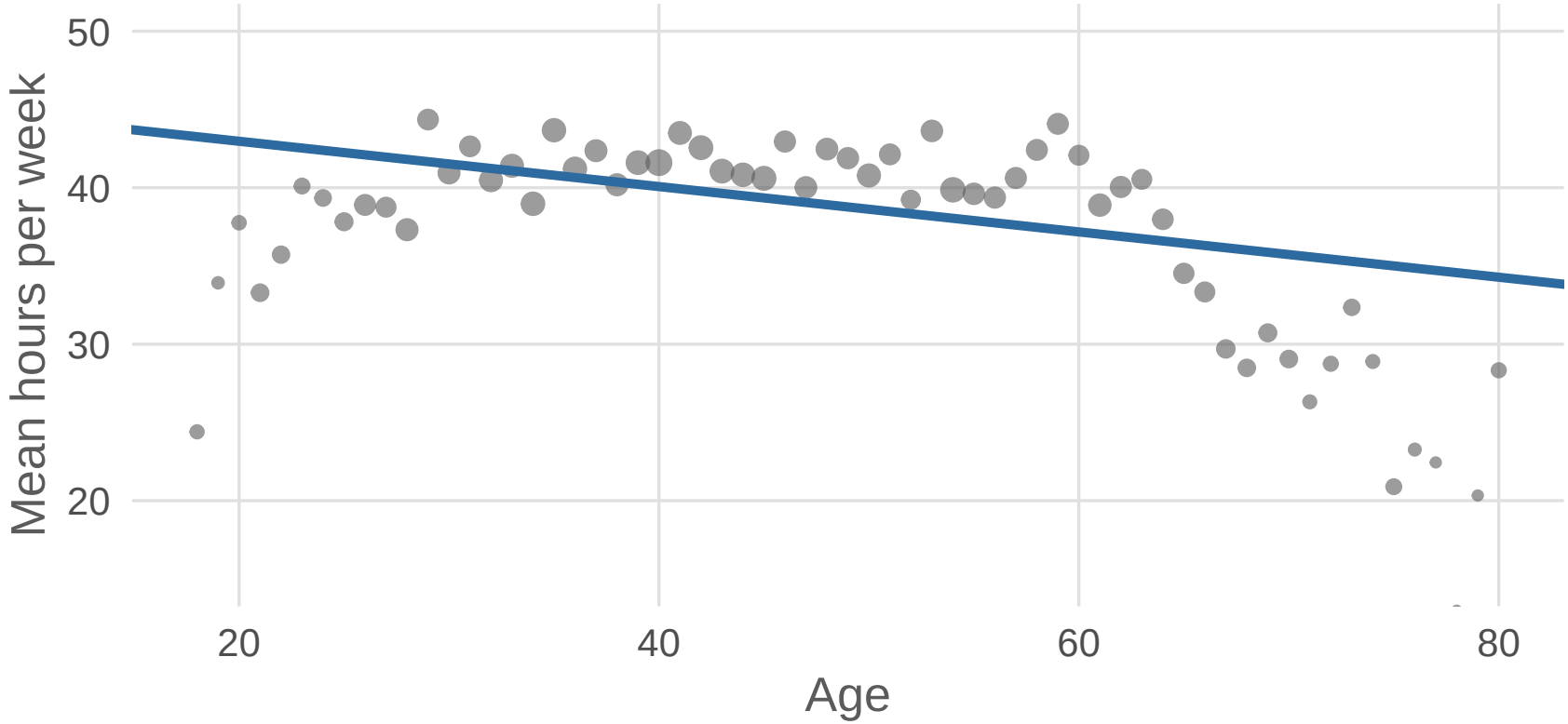
Interactions

Polynomials

Adjusted R^2

Hours worked and age

ANES respondents who worked in the past 12 months: usual hours per week, averaged at each age.



The means rise until about age 30. They stay near 41 hours until about age 60. After 60, they fall. A straight line has one slope at every age.

Linear regression can be used to fit a curve

Add age squared as a second variable:

$$\texttt{hours} = \beta_0 + \beta_1 \texttt{age} + \beta_2 \texttt{age}^2 + \varepsilon$$

- The model is linear in the coefficients: each β multiplies a variable.
- $\texttt{age}^2$ is a new variable, computed from `age`.

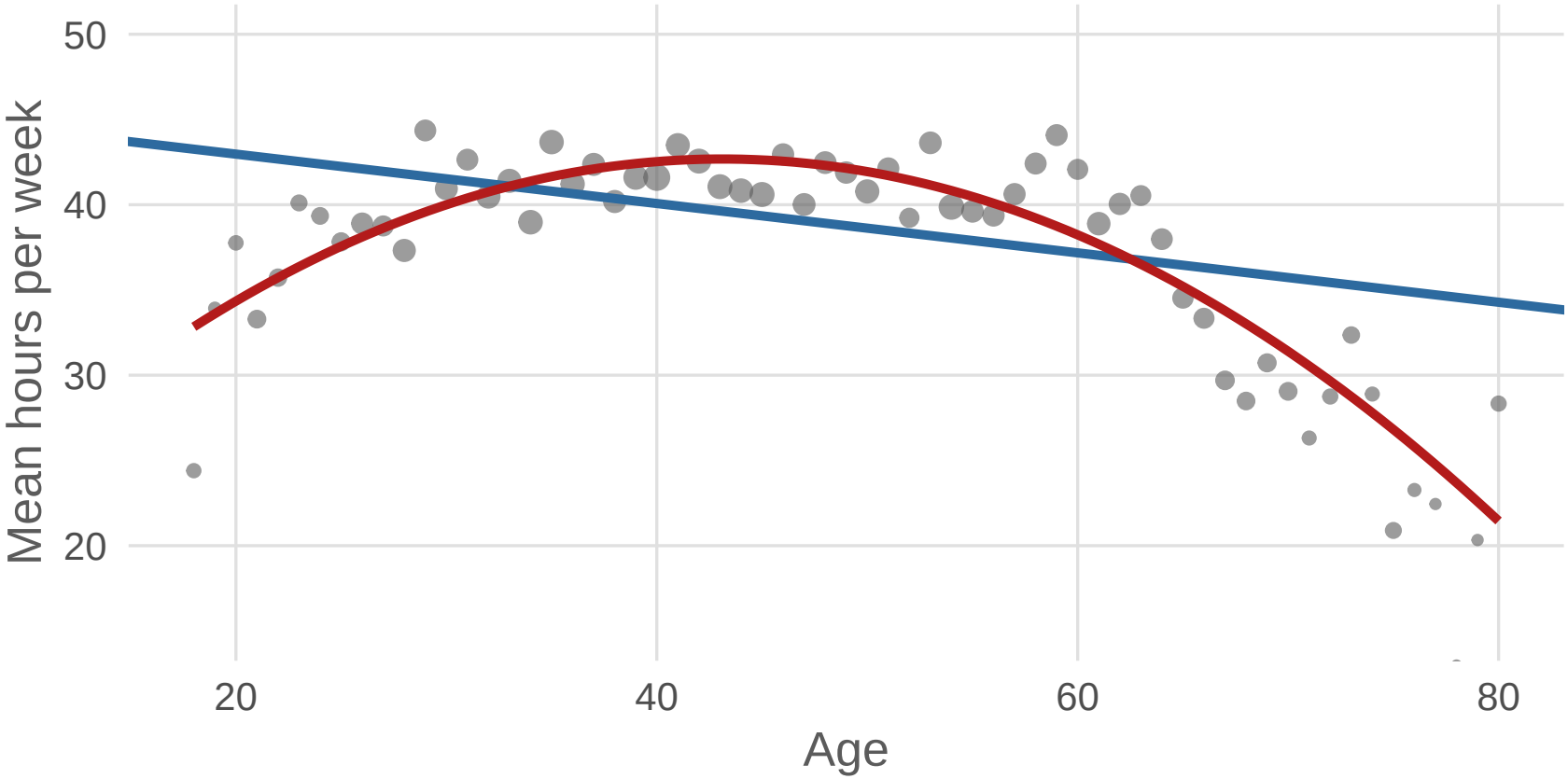
```
fit_h2 <- lm(hours ~ age + I(age^2), data = work)
```

```
(Intercept)         age    I(age^2)
    13.6717      1.3457     -0.0156
```

`I()` tells R to compute `age^2` as arithmetic.

$$\widehat{\texttt{hours}} = 13.7 + 1.35\,\texttt{age} - 0.016\,\texttt{age}^2$$

The fitted curve



$$\widehat{\texttt{hours}} = 13.7 + 1.35\,\text{age} - 0.016\,\text{age}^2$$

The slope changes with age

The fitted curve:

$$\widehat{\texttt{hours}} = \hat{\beta}_0 + \hat{\beta}_1 \, \texttt{age} + \hat{\beta}_2 \, \texttt{age}^2$$

The slope of the curve at age a is the derivative of $\widehat{\texttt{hours}}$ with respect to age:

$$\frac{d\,\widehat{\texttt{hours}}}{d\,\texttt{age}} = \hat{\beta}_1 + 2\hat{\beta}_2 \, a = 1.35 - 0.031\,a.$$

Age	25	43	65
Slope (hours per year of age)	0.57	0.00	-0.68

- The slope is 0 at the peak: $a = -\hat{\beta}_1/(2\hat{\beta}_2) = 43.1$.
- Neither $\hat{\beta}_1$ nor $\hat{\beta}_2$ alone is the marginal effect of age. As with an interaction, we report the slope at chosen ages.

Roadmap

Where we are

Omitted variable bias

Three routes to a zero conditional mean

Dummy variables

Interactions

Polynomials

Adjusted R^2

Adding a variable never lowers R^2

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}$$

Regression	R^2
`hours ~ age`	0.021
`hours ~ age + I(age^2)`	0.080

- The OLS coefficients minimize the sum of squared residuals.
- With one more variable, a coefficient of 0 on that variable reproduces the old fit. So the minimum sum of squared residuals cannot rise.
- So R^2 cannot fall. R^2 rises even when the new variable is pure noise.

Adjusted R^2

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

k is the number of explanatory variables. Adjusted R^2 equals R^2 minus a penalty that grows with k.

Regression	n	k	R^2	Adjusted R^2
Hours on age, age^2	3,374	2	0.080	0.079
`u5mort` on `goveff`, log GDP p.c.	46	2	0.44	0.41

The adjustment is small when n is large relative to k.

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, sections 2.3.7, 6.7 and 7.5.
- Aronow and Miller, *Foundations of Agnostic Statistics*, sections 4.1–4.2 and 4.4.
- Arel-Bundock, *Model to Meaning*, "Slopes."
- Brambor, Clark and Golder (2006), "Understanding interaction models: improving empirical analyses," *Political Analysis* 14(1).
- Angrist and Pischke (2009), *Mostly Harmless Econometrics*, chapter 3.
- Angrist (2015), "regression: BLP not BLUE!", *Mostly Harmless Econometrics* blog, mostlyharmlesseconometrics.com/?p=1078.
- Buja et al. (2019), "Models as approximations I: consequences illustrated with linear regression," *Statistical Science* 34(4).
- **`handout_multiple_regression.pdf`**: multiple regression in matrix form.
- **`handout_ovb.pdf`**: quantifying omitted variable bias.

These slides draw on teaching materials by Matt Blackwell, Amitabh Chandra, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto. Data: World Bank, World Development Indicators and Worldwide Governance Indicators; American National Election Studies 2024.